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Synthesis, Exfoliation, and Transport Properties of Quasi-1D van der Waals Fibrous Red Phosphorus

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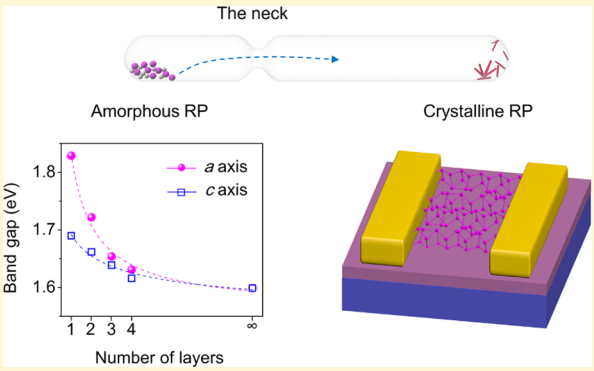
ABSTRACT: Quasi-one-dimensional (quasi-1D) fibrous red phosphorus (RP) has triggered growing interest recently for its unique properties and promising applications in energy storage, sensing, and heterogeneous catalysis. However, attempts to explore its anisotropy and transport properties have progressed slowly. In this work, we report the fabrication of large-size and pure bulk fibrous RP crystals using a chemical vapor transport reaction by modifying growth kinetics. For the first time, the weak layer number-dependent anisotropic electronic structure of fibrous RP is revealed based on theoretical calculations. The *b*-axis-aligned fibrous RP nanoribbons with a high aspect ratio exceeding 1000 are easily prepared by facile liquid exfoliation. Impressively, the field-effect transistor device built with fibrous RP nanoribbons exhibits p-type transport behavior and a high mobility reaching 236.7 $cm^2 V^{-1} s^{-1}$ with an On/Off ratio approaching 1.6×10^3, which are superior to those of other reported RP-based materials in general. The results demonstrate the great potential of fibrous RP as a promising channel material and lay a solid foundation to further study the intrinsic properties of quasi-1D van der Waals materials.



INTRODUCTION

Quasi-one-dimensional (quasi-1D) materials have attracted tremendous interest recently for marvelous physical properties and emerging applications in next-generation electronic and optoelectronic devices.[1,2] Generally, bulk quasi-1D materials are composed of parallel-arranged atomic chains weakly bound in bundles by van der Waals forces. The unique chain-like structure gives rise to fantastic physical properties, such as topological superconductivity,[3] charge density waves,[4] topological insulation,[5] and superconductor−insulator transitions.[6] Moreover, the inherent strong structure anisotropy of quasi-1D materials makes them desirable for the detection of polarized light and polarization-sensitive photodetectors.[7,8] Compared with two-dimensional (2D) materials, not only the thickness but also the width could be adjusted for quasi-1D materials, offering another degree of freedom for tailor-made properties to satisfy various application requirements. Consequently, the explorations of emerging quasi-1D materials are highly desirable, especially controllable synthesis and systematic investigation of the carrier transport properties of these materials.

As one well-known allotrope of elemental phosphorus, red phosphorus (RP) is much abundant, easily available at low cost, chemically stable, and environmentally friendly. Nowadays, it is widely applied to igniters, flame retardants, pesticides, and chemical analysis in our daily life and traditional industry.[9] Recently, RP has aroused the interest of a growing number of chemists and material scientists because it presents some fascinating properties which may find applications in semiconductors, rechargeable batteries, and photocatalysts.[10,11] RP modifications exist in five distinct allotropes, namely Type-I(amorphous RP), Type-II, Type-III, Type-IV(fibrous RP), and Type-V(violet RP or Hittorf's phosphorus).[12] Among these, fibrous RP is an emerging quasi-1D material composed of strong intratube covalent bonds and weak intertube van der Waals forces.[13] The unique quasi-1D structure and the associated promising properties of fibrous RP render its potential applications in electrochemical nitrogen fixation,[14] photocatalytic CO_2 reduction,[15,16] sonothermal therapy,[17] and live-cell imaging.[18,19] Additionally, Nilges' group reported the synthesis of bunch-like crystalline fibrous RP employing $CuCl_2$ as a mineralizer for the chemical vapor transport (CVT) reaction.[20] In a study by Ji et al. in 2016, ultra-long crystalline RP nanowires were obtained and their field-effect transistor (FET) mobility reached 300 $cm^2 V^{-1} s^{-1}$ with an On/Off ratio approaching 10^2. However, the obtained nanowires were composed of RP mixtures, and the intrinsic electronic performance of pure fibrous RP was not explored.[21] Zhu et

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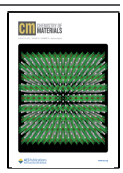

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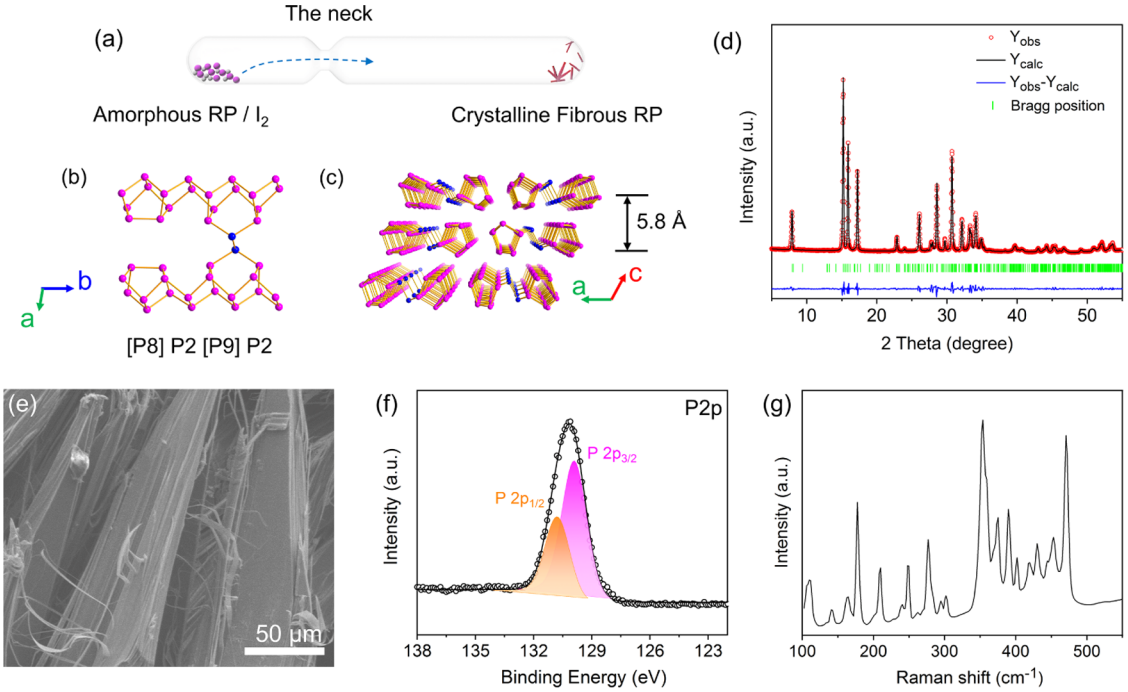

Figure 1. Synthesis of bulk crystalline fibrous RP using the CVT method. (a) Schematic illustration of preparation route for bulk crystalline fibrous RP via the CVT reaction. (b) Building units of the tubes in fibrous RP. Blue atoms represent the linkages of double tubes. (c) Lattice arrangement of fibrous RP viewed along the *b*-axis. (d) Lebail refinement profiles of bulk fibrous RP. Red circles and the black line denote the observed and calculated XRD patterns, respectively. The blue line indicates the difference profile. The green sticks mark the position of the reflections for fibrous RP. The refinement convergence was achieved with R_{wp} = 7.72 and χ^2 = 8.48. (e) Scanning electron microscopy (SEM) image, (f) fine X-ray photoelectron spectroscopy (XPS) spectrum P 2p, and (g) Raman spectrum of bulk fibrous RP.

al. prepared urchin-like fibrous RP using the iodine-assisted CVT method and investigated its lithium storage performance.[22] Son's group first created a solution-processed method for the fabrication of fibrous RP thin films. Unfortunately, the process involves tedious synthesis procedures, most of which have to be carried out under nitrogen or inert gas protected conditions. The carrier mobility of the fibrous RP thin films was only 1.8×10^{-6} cm^2 V^{-1} s^{-1} due to K-doping and low crystallinity.[23] Further exploration on physical anisotropic properties and intrinsic carrier transport performance, which are the crucial characteristics for quasi-1D materials, hinges on the advances in the synthetic strategy toward high-quality fibrous RP bulk crystals. However, the abovementioned synthesis strategies mainly result in, bunches-like,[20] urchin-shaped fibrous RP crystals,[22] or even phase-impure RP.[21] Till date, the fabrication of large-sized, pure and bulk crystalline fibrous RP suitable for exfoliation of high-aspect-ratio fibrous RP is rarely reported, thereby hampering the systematic study on its anisotropic properties and intrinsic carrier transport performance.

Herein, we report the successful preparation of large-sized fibrous RP with high crystallinity by modifying growth kinetics in the CVT reaction. For the first time, our theoretical calculation reveals the anisotropic electronic structure and the weak layer number-dependent band gap of fibrous RP. A definitive correlation of exfoliation efficiency with solvent surface tensions is established for liquid exfoliation of fibrous RP. The *b*-axis-aligned fibrous RP nanoribbons with a high aspect ratio exceeding 1000 are first prepared by facile liquid exfoliation. More importantly, the FET device made with fibrous RP nanoribbons exhibits p-type transport behavior and an impressively high mobility reaching 236.7 cm^2 V^{-1} s^{-1} with an On/Off ratio approaching 1.6×10^3, superior to that of other reported RP-based materials.

■ RESULTS AND DISCUSSION

Synthesis and Structural Analysis. CVT is a traditional crystal growth technique which could date back to the 1850s.[24] A representative CVT reaction involves three major processes, including (1) sublimation and reaction of precursors with transport agents at the source region, (2) gaseous intermediate movement, and (3) crystallization at the growth region. To acquire the bulk crystalline fibrous RP instead of nanostructured RP such as bunches-like,[20] nanowires,[21] and urchin-shaped,[22] it is crucial to dramatically slow down the growth rate. Generally, based on intensive studies on the growth kinetics of CVT, the growth rate is controlled by gas transport, including convection and diffusion, which are the rate-determining steps in three major processes.[25,26] Thus, the crux is to slow down gas convection and diffusion. The uniform temperature strategy by using a muffle furnace is employed to realize low gas convection mostly resulted from temperature gradient between source and growth regions. Moreover, the gas diffusion rate can be described by Schäfer's transport equation[27]

$$J \propto \frac{p_i}{\sum p} \cdot \frac{s}{l} \cdot \overline{T}^{0.75}$$

The p_i and $\sum p_i$ are the partial pressure of the transport effective species and total pressure, respectively; s and l are cross-section and length of the quartz ampoule, respectively; and $\overline{T}$ is the mean temperature. Therefore, to slow down the gas diffusion rate, we should minimize the amount of RP, increase the length of ampoule, and lower mean temperature

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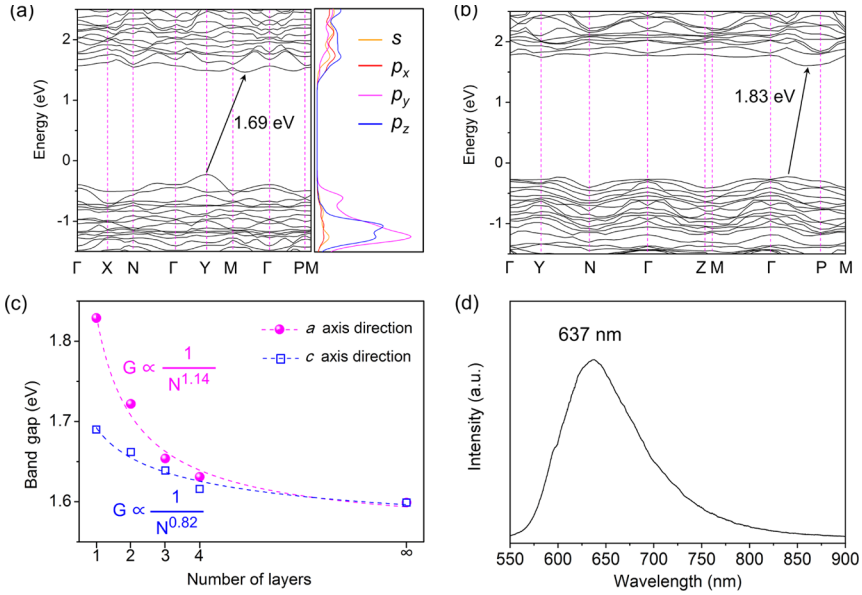

Figure 2. Band structure of fibrous RP. The calculated electronic structures of monolayer fibrous RP (a) along the *c*-axis and (b) along the *a*-axis. The right insert in (a) shows the orbital-resolved electronic density of states. (c) Thickness-dependent band gaps of fibrous RP obtained from GGA-PBE calculation. (d) PL spectrum of bulk fibrous RP showing an optical PL emission peak at about 637 nm.

moderately. In addition, a "neck," as shown in Figure 1a, is designed on the quartz ampoule to decrease the cross-section area, thereby slowing down the reaction rate.

Based on the above discussion, we achieve the successful preparation of large-sized bulk crystalline fibrous RP by modifying growth kinetics in the CVT reaction, as schematically illustrated in Figure 1a. In this work, commercial amorphous RP (Figure S1) and iodine were selected as the precursor and transport agent, respectively. The detailed growth procedure can be found in the Supporting Information.

Fibrous RP is an emerging quasi-1D material composed of strong intratube covalent bonds and weak intertube van der Waals forces. The building block is [P8]P2[P9]P2, denoting that [P8] cage and [P9] cage are connected by the P2 bridge in the tube. Additionally, every two adjacent tubes are chemically bonded by the linkages, shown as the blue atoms in Figure 1b. Different from 2D-layered material, double tubes in fibrous RP are stacked by van der Waals forces both along the *c*-axis and *a*-axis (Figure 1c). Consequently, novel fundamental science and applications ranging from nanoelectronics to composites could be expected for fibrous RP considering its unique structural characteristics.

The powder X-ray diffraction (XRD) pattern is well fitted in the triclinic cell with lattice parameters a = 12.1729(8) Å, b = 13.0042(9) Å, c = 7.1369(6) Å, α = 117.90(5) °, β = 106.19(2) °, and γ = 97.40(4) ° using the Lebail refinement method (Figure 1d, see Supporting Information for more details). This refinement confirms the pure single phase fibrous RP structure, and the sharp diffraction peaks denote high crystallinity. The SEM images (Figures 1e, S2a,b) show clearly regular boundaries and edges of bulk fibrous RP. The elemental mapping and energy-dispersive spectra (EDS), as shown in Figure S2c,d, exhibit pure P element composition. Moreover, XPS analysis, as shown in Figures S3 and 1f, presents the $2p_{3/2}$ and $2p_{1/2}$ doublet at 129.9 and 130.8 eV, further confirming the only existence of elemental P without other valence states. The typical sharp Raman peaks, as shown in Figure 1g, indicate the formation of well-crystalline fibrous RP.[18] Additionally, given diverse atomic coordination in such a low-symmetric triclinic unit cell, it is not easy to accurately assign the Raman peak to each individual vibrational mode.

Anisotropic Electronic Structure. The anisotropic electronic structure of fibrous RP from monolayer to bulk was investigated using the density functional theory (DFT) method. It is observed that monolayer fibrous RP along the *c*-axis is an indirect band gap semiconductor with a band gap of 1.69 eV, of which the valence band maximum (VBM) and conduction band minimum (CBM) locate at the Y point and between M and Γ points, respectively (Figure 2a). Moreover, to provide further insights into the electronic structure of monolayer fibrous RP, orbital-resolved density of states was calculated. As presented in the insert in Figure 2a, the VBM is mostly dominated by p_y orbital, and the CBM is almost equally formed by p_x and p_z orbitals. For thicker fibrous RP along the *c*-axis, as depicted in Figures 2c and S4, the band gap ranges from 1.69 eV (monolayer) to 1.60 eV (bulk). The values are fitted by power law as following equation[28]

$$G = \frac{A}{N^{\alpha}} + B$$

where G is a band gap value, N is layer numbers, and A and B are constant. This fit follows the $1/N^{0.82}$ power law, suggesting extraordinarily slow band gap decay. This evidence reveals that the band gap value is not susceptible to the number of layers, or in other words, it is difficult to modulate band gap significantly by just altering the number of layers. These evidence demonstrate the lack of strong interlayer interactions for fibrous RP along the *c*-axis.[29] Besides the *c*-axis, a van der Waals interaction also exists along the *a*-axis. The larger band gap of 1.83 eV along the *a*-axis can be observed from Figure 2b. As shown in Figure 2c and S5, the band gap gradually decreases from 1.83 eV for monolayer to 1.60 eV for bulk along the *a*-axis and follows $1/N^{1.14}$ power law. Above results demonstrate that a stronger van der Waals interaction exists along the *a*-axis compared to the *c*-axis. These results also imply that fibrous RP nanoribbons or nanosheets along the *c*-

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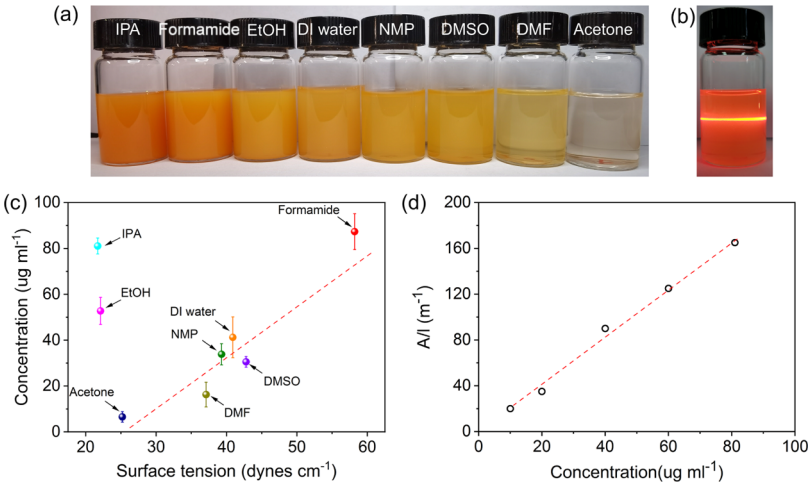

Figure 3. Liquid exfoliation of bulk fibrous RP. (a) Photograph of dispersions for exfoliated fibrous RP nanoribbons in different solvents. (b) Tyndall effect of fibrous RP dispersion in NMP. (c) Dispersion concentration plot for various solvents with different surface tensions after 3000 rpm centrifugation for 30 min. (d) Absorbance (A, at λ = 650 nm) divided by cell length (l) as a function of the concentration of dispersed fibrous RP nanoribbons in IPA.

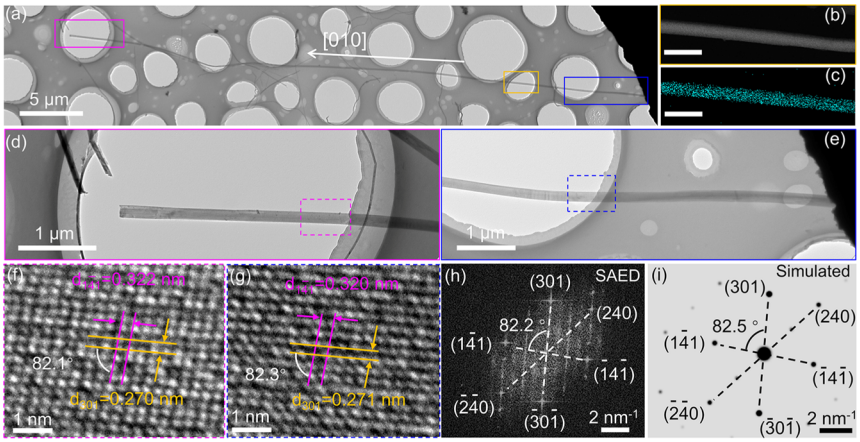

Figure 4. TEM characterization of fibrous RP nanoribbons. (a) Representative low-magnification image of fibrous RP nanoribbons acquired by liquid exfoliation in IPA. The white arrow indicates the [010] crystallographic direction of the nanoribbon. (b) HAADF image of the nanoribbon taken from the selected area [outlined in orange rectangular in (a)]. The scale bar is 250 nm. (c) Corresponding element mapping of P. Scale bar is 250 nm. (d) Enlarged TEM image taken through the pink box in (a). (e) Enlarged TEM image taken from the blue box in (a). (f) (g) Lattice fringes of the nanoribbon marked by pink dashed rectangular in (d) and blue dashed rectangular in (e), respectively. (h) Electron diffraction pattern obtained from FFT according to Figure 4f using Gatan Digital Micrograph software. (i) Simulated electron diffraction pattern viewed along the zone axis [4 $\bar{2}\bar{1}\bar{2}$] for fibrous RP using Crystal Maker software.

axis could be fabricated more easily via exfoliation. Photoluminescence (PL) and ultraviolet−visible−near infrared (UV−vis−NIR) diffuse reflectance spectra were measured to determine the band gap of bulk fibrous RP experimentally. PL spectrum, as shown in Figure 2d, demonstrates an optical PL emission peak at about 637 nm (1.95 eV). The difference compared with calculated band gap (1.60 eV) is ascribed to the inherent underestimation for GGA-PBE functional.[28] Moreover, as presented in Figure S6a, bulk fibrous RP exhibits excellent and wide light absorption capability from UV to NIR regions. The estimated band gap is 1.86 eV from a Tauc plot (Figure S6b). The interband transition from the defect energy level to conduction band would lead to the tailing effect in UV−vis absorption spectrum; thus, band gap redshift occurs. As a result, the band gap (1.86 eV) obtained from the Tauc plot is smaller than that (1.95 eV) from the PL spectrum.

Exfoliation Investigation. Solvent-phase exfoliation is probably the most practical route to achieve scalable production of high-aspect-ratio fibrous RP nanoribbons. Previous studies prove that the choice of solvents is decisive.[30] Thus, we investigated the exfoliation efficiency of eight solvents with various surface tensions, including isopropanol (IPA), formamide, ethanol (EtOH), deionized water, *N*-methyl-2-pyrrolidone (NMP), dimethyl sulfoxide, dimethylformamide, and acetone. The photograph, as shown in Figure 3a, exhibits extremely different exfoliation efficiency for various solvents. In addition, the Tyndall effect can be observed in NMP, confirming the formation of stable dispersion composed of nano-scale materials (Figure 3b). The stable dispersions possess fairly excellent light-harvesting capability, as can be seen from the UV−vis−NIR absorption spectra (Figure S7).

To determine the precise concentration of the suspension, inductively coupled plasma-atomic emission spectroscopy (ICP-AES) analysis was performed. The plot of dispersion concentration versus surface tension demonstrates that there is a broadly linear relationship (Figure 3c). The solvents with

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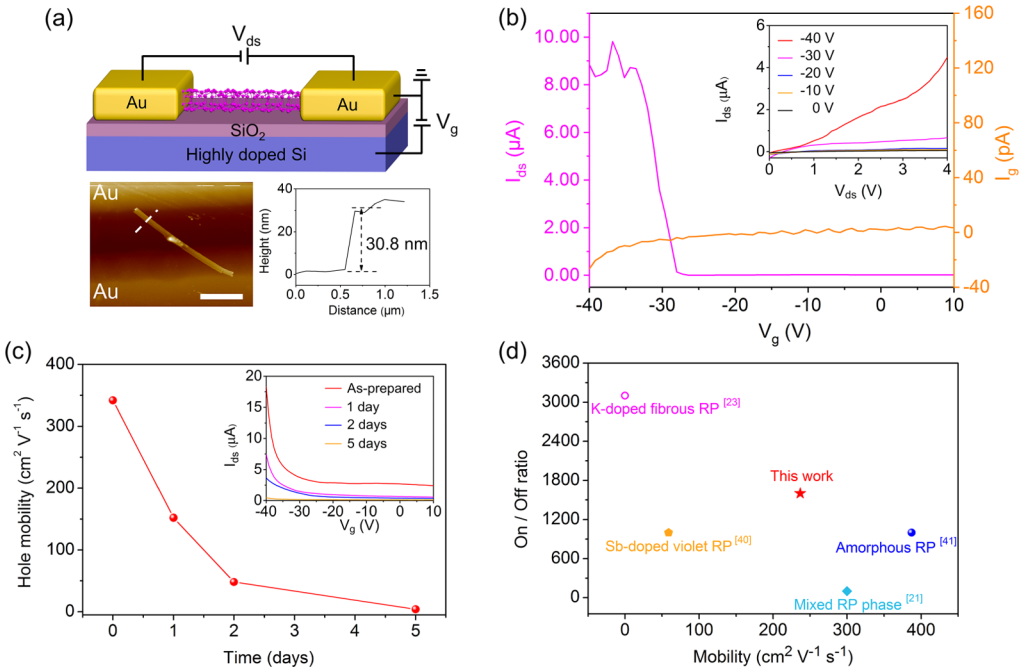

Figure 5. Electrical transport of the fibrous RP nanoribbons. (a) Top, schematic diagram of a FET based on fibrous RP nanoribbons. Bottom left, atomic force microscopy (AFM) of a typical FET device. The scale bar is 5 μm. Bottom right, height of the fibrous RP nanoribbon. (b) Transfer curves of the FET utilizing the exfoliated 30.8 nm thick fibrous RP nanoribbon as a channel material with drain-source voltages of 8 V. The insert shows the corresponding output curves with V_g varying from −40 to 0 V. I_{ds}, V_{ds}, V_g, and I_g are the source-drain current, source-drain bias, back gate bias, and gate leakage current, respectively. (c) Stability evolution of the fibrous RP nanoribbon-based FET device. The insert is the transfer curve of the FET device after storage for different times. (d) Comparison of FET device performance for the reported RP materials.

higher surface tension deliver high exfoliation efficiency due to strong phosphorus-solvent adhesion which could effectively stabilize the newly exposed nanoribbon surfaces. The tendency is similar to liquid exfoliation of transition-metal dichalcogenides.[31] The anomalous highly efficient exfoliation for IPA and EtOH with low surface tension is presumably interpreted by its planar molecular structure. Specifically, the planarity structure of IPA on the leading edge could increase the sharpness of the wedge, allowing for the enhanced penetration of solvent molecules into the interlayer gaps and increasing the stabilization of freshly exposed surfaces.[32] Even though formamide possesses the highest exfoliated concentration, IPA is still regarded as the optimal solvent considering its low-boiling-point, cheap, easy-to-handle, and low-toxicity characteristics. Besides, IPA also achieves a considerable exfoliation yield of 16%. Moreover, the extinction coefficient for IPA dispersion is determined to be about α = 2087 L g^{-1} m^{-1}, according to Lambert−Beer's law at λ = 650 nm (Figure 3d).

The morphology and lattice orientation of fibrous RP nanoribbons obtained from liquid exfoliation in IPA were further characterized thoroughly. A high-aspect-ratio nanoribbon of roughly 45 μm in length can be observed in the representative transmission electron microscopy (TEM) image, as shown in Figure 4a. The local high-angle annular dark field and accessorial EDS mapping image, as presented in Figure 4b,c, confirm that the nanoribbon is only composed of P element. The magnified images, as shown in Figure 4d,e, reveal regular edges clearly. To unveil the lattice orientation of nanoribbons, high-resolution TEM and corresponding fast Fourier transform (FFT) were conducted. High-resolution TEM is taken at two different locations marked by pink and blue dashed rectangular along this nanoribbon. As shown in Figure 4f, the observed highly ordered lattice fringes denote high crystallinity. The d-spacings of 0.322 and 0.270 nm are visible, corresponding to the $(1\bar{4}1)$ and (301) planes, respectively. The angle between the two planes is measured to be 82.1°, nearly same as the theoretically calculated value 82.5°. The other end of the nanoribbon displays similar lattice fringe (Figure 4g). Therefore, the fibrous RP nanoribbon is *b*-axis-aligned along the length, namely [010] direction according to crystallographic theory (as indicated by the white arrow in Figure 4a). The FFT (Figure 4h) is well in line with the simulated electron diffraction pattern, as shown in Figure 4i, also completely demonstrating its crystallographic [010] orientation. The size distribution of fibrous RP nanoribbons in IPA is shown in Figure S8a−c, acquired from more TEM images in Figure S9. Compared to other solvents (Figure S10), the optimal exfoliation capability was achieved in IPA. The average length is 26 μm, and the average width is about 70 nm. The highest aspect ratio could exceed 1000, which is far larger than the previous reports for phosphorus nanoribbons.[14,33] Such a facile liquid exfoliation capable of forming *b*-axis-aligned fibrous RP nanoribbons with high aspect ratio is easily accessible to most research laboratories. It may boost more extensive exploration on its properties and applications.

Carrier Transport Performance. To study the intrinsic carrier transport property of quasi-1D fibrous RP, an FET was made based on a fibrous RP nanoribbon. A completely dry heat-release-tape-assisted mechanical exfoliation technique was employed in the glovebox to avoid oxidation and lattice defects possibly induced by liquid exfoliation. Figure S11a presents the schematic diagram, and detailed procedures can be found in Supporting Information. For 2D materials like graphene and black phosphorus with relatively large contact areas, a common Scotch tape is suitable to achieve exfoliation. However, it's not feasible for quasi-1D materials like fibrous RP. The heat-

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release-tape-assisted mechanical exfoliation technique, in which the adhesion force of the tape could disappear at a specific temperature, can enable a high exfoliation efficiency.[34] The optical micrograph, as shown in Figure S11b, displays the morphology of one exfoliated fibrous RP nanoribbon. The Raman peak of exfoliated nanoribbon remains nearly the same as that of bulk fibrous RP, confirming that the exfoliation procedure is nondestructive (Figure S12). The XRD peaks, as shown in Figure S11c, are dominated by (001) and (002) reflections, revealing highly preferred orientation. Moreover, most exfoliated nanoribbons are stacked along the *c*-axis due to a weaker interlayer interaction compared to the *a*-axis, which is consistent with the aforementioned theoretical calculation results. The height is measured to be 10.5 nm, as shown in Figure S11d.

A bottom-gated FET utilizing a mechanically exfoliated fibrous RP nanoribbon as the channel material was fabricated on a highly doped Si substrate covered with a 300 nm thick thermally oxidized silicon dioxide layer. Figure 5a displays a schematic diagram of the device structure. A facile fabrication technique, namely, the Cu grid mask method,[35] was adopted to fabricate the bottom-gated FET. As depicted in Figure 5b, the source-drain current (I_{ds}) decreases with the back gate bias (V_g) varying from −40 to 10 V, suggesting a typical p-type transport behavior. In addition, the nonlinear characteristic of output curves (insert of Figure 5b) demonstrates the typical Schottky contact.[36] The field-effect mobility was extracted using the following equation

$$u = \frac{dI_{ds}}{dV_g} \times \frac{L}{WC_iV_{ds}}$$

where dI_{ds}/dV_g is the slope of the linear regime of transfer curves, C_i is the dielectric capacitance between the channel and the back-gate per unit area, L and W are the channel length and width between source/drain electrode, respectively, and V_{ds} is the source-drain bias. Here, it exhibits a high hole mobility, reaching up to 236.7 $cm^2\ V^{-1}\ s^{-1}$. The value, however, is far less than the theoretically predicted value of $1.77 \times 10^4\ cm^2\ V^{-1}\ s^{-1}$.[37] This might be caused by large contact resistance between channel materials and electrodes. The fact implies that higher mobility could be achieved by altering the contact electrode or employing high-k materials such as HfO_2 as a gating layer. The feasibility of this strategy has been proved for MoS_2.[38] As shown in Figure S13, a relatively high On/Off ratio of about 1.6×10^3 can be observed. Moreover, the lower leakage current (I_g) can largely reduce power consumption.

The stability of fibrous RP nanoribbon-based FETs with a thickness of 59.3 nm (Figure S14) was investigated for the first time, which were exposed to the dark environment (a relative humidity of ∼31% and temperature of 16 °C). Figure 5c and the insert demonstrate the hole mobility evolution with increasing exposure time. After 1 day's exposure, hole mobility decreases dramatically, indicating suppressed carrier transport due to surface oxidation. Current modulation capability would disappear after 5 days' exposure. These evidence demonstrate fairly low stability of fibrous RP nanoribbon, which is most likely caused in the lone pairs existing in fibrous RP. Similar to black phosphorus nanosheets, physical encapsulation, surface chemical modification, or doping may effectively strengthen its ambient stability.[39] Figure 5d and Table S1 exhibit the comparison of transport performance of our pure fibrous RP nanoribbons and other reported RP materials. The mobility of fibrous RP nanoribbons is much superior to K-doped fibrous RP[23] with low crystallinity obtained from the solution-processed reaction. Moreover, compared to Sb-doped violet RP,[40] RP nanowires with mixed phase,[21] and amorphous RP film,[41] pure fibrous RP nanoribbons possess a higher On/Off ratio. The characteristics of fairly high mobility and On/Off ratio render their potential applications as logic devices.

■ CONCLUSIONS

In summary, large-sized bulk fibrous RP with high crystallinity was synthesized by optimizing growth kinetics in the CVT reaction. DFT calculation reveals the anisotropic electronic structure and the weak layer number-dependent band gap of fibrous RP. The weak interaction between layers along the *a*-axis and *c*-axis facilitates the liquid exfoliation of fibrous RP bulk crystals. For the first time, the *b*-axis-aligned fibrous RP nanoribbons with a high aspect ratio exceeding 1000 are prepared, and the extinction coefficient in IPA is determined to be about α = 2087 L g^{-1} m^{-1}. A proportional correlation between exfoliation efficiency of fibrous RP nanoribbons and solvent surface tension was established. The transport performance of fibrous RP nanoribbons was investigated, and it demonstrates p-type behavior and remarkable hole mobility reaching 236.7 $cm^2\ V^{-1}\ s^{-1}$ with an On/Off ratio of 1.6×10^3. The results not only demonstrate the great potential of fibrous RP nanoribbons as a promising channel material but also offer a facile practicable fabrication route of quasi-1D phosphorus nanoribbons.

■ EXPERIMENTAL SECTION

Growth of Bulk Crystalline Fibrous RP. Bulk crystalline fibrous RP was synthesized by a CVT reaction using amorphous RP as the precursor and iodine (I_2) as the transport agent. 200 mg of amorphous RP lump (Alfa Aesar, 99.999%) and 10 mg of I_2 (Alfa Aesar, 99.9%) were sealed under vacuum of 10^{-3} pa in a quartz ampoule (inner diameter of 1.0 cm and ampoule length of ∼15 cm). Especially, a neck of 0.4 cm inner diameter was designed on the quartz ampoule to modulate transport dynamics (Figure 1a). The sealed ampoule was then placed horizontally in the muffle furnace. The furnace was heated to 600 °C at a heating rate of 5 °C/min and maintained for 2 h followed by cooling at a rate of 2 °C/min to room temperature. Bulk fibrous RP was obtained at the growth side after breaking the quartz ampoule.

Characterization of Fibrous RP. *XRD and Lebail Refinement.* The XRD patterns were collected on a Bruker D8 ADVANCE diffractometer with Cu Kα radiation (λ = 1.5406 Å) at 40 kV and 40 mA. X-ray data refinement was carried out with a Lebail refinement method using Fullprof software. The background, zero point, lattice parameters, and coefficients for the peak shape function were refined until convergence was achieved with R_{wp} = 7.72 and χ^2 = 8.48. The data could be fitted in the triclinic cell with lattice parameters a = 12.1729(8) Å, b = 13.0042(9) Å, c = 7.1369(6) Å, α = 117.90(5)°, β = 106.19(2)°, and γ = 97.40(4)°.

Optical Micrograph and SEM. Optical micrograph (OM) was obtained by an optical microscope (OLYMPUS BX51) at 50× magnification. The morphology of bulk fibrous RP was identified by SEM (Hitachi SU8010) at 15 kV equipped with an energy-dispersive X-ray spectroscope.

Raman Spectra and PL. The PL and Raman spectra of fibrous RP were collected on a high-resolution confocal Raman system (HORIBA Jobin Yvon LabRAM HR Evolution) equipped with a 532 nm laser as the excitation source at room temperature. The beam spot diameter of the laser was about 1.5 μm, and a 1% laser power cutoff was chosen.

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TEM and High-resolution TEM. The low-magnification TEM images were obtained with H-7650B at 80 kV. The lattice fringe images of fibrous RP nanoribbons were recorded by high-resolution TEM (JEOL JEM 2100F).

XPS and AFM. The valence of P was identified by using X-ray photoelectron spectra (PHI Quantera) with monochromatic Al Kα radiation (hν = 1486.7 eV). AFM (Bruker Dimension Icon) image was recorded in a tapping mode to observe the mechanically exfoliated fibrous RP nanoribbons on SiO_2/Si wafers.

Absorption Spectrum and Diffuse Reflectance Spectrum. Optical absorption spectra of fibrous RP nanoribbon dispersion were acquired using a Shimadzu UV-2600 spectrophotometer. A quartz cuvette with 10 mm path length was used for the measurements. The baseline for various solvents was subtracted from the spectra. The UV−vis−NIR diffuse reflectance spectra were also performed on a Shimadzu UV-2600 spectrophotometer equipped with an integrating sphere. Due to its almost 100% reflectivity, $BaSO_4$ was chosen as a reference sample.

Liquid Exfoliation and Suspension Concentration Determination. For typical liquid exfoliation experiments, 10 mg of bulk fibrous RP was added into 20 mL of IPA in the vial. To prevent ambient exposure, vial was filled with argon and sealed with parafilm tightly, then the vial was ultrasonicated at a bath ultrasonic power of 100 W (KQ2200DB, Kunshan Ultrasonic Instruments) for 30 min. The small broken fibrous RP flakes rather than nanoribbons were obtained if the ultrasonic time exceeded 30 min. Bath water maintained a temperature between 20 and 30 °C using an ice bag. After exfoliation, the suspensions were centrifuged at 3000 rpm for 30 min to discard bulk crystals. Similar processes were also performed for other solvents.

For low concentration solutions, Lambert−Beer's law is usually used to calculate the concentration of solutions. However, owing to the significant scattering for nanoribbons or flakes, maybe it is not appropriate to measure the concentration of suspensions based on Lambert−Beer's law. Thus, the fibrous RP nanoribbons in the suspension were decomposed by nitric acid first. Then, the exact concentration of the suspension in this work was determined by ICP-AES (iCAP6300, Thermo Fisher).

Heat Release Tape-Assisted Mechanical Exfoliation for Fibrous RP Nanoribbons. Compared to 2D nanosheets with a relatively large area, 1D nanoribbons were hardly exfoliated and transferred on the substrate using a common scotch tape. Therefore, the heat release tape-assisted mechanical exfoliation technique enabling high exfoliation efficiency for nanoribbons was performed. In brief, first, the as-synthesized bulk fibrous RP was exfoliated repeatedly by using heat release tapes. Then, the tape with fibrous RP nanoribbons was transferred to the SiO_2/Si substrate. Subsequently, the substrate was placed on a hot plate at 110 °C for 30 s. Once heated beyond the release temperature, the tape will lose its adhesion and fall off automatically. Finally, the fibrous RP nanoribbons on the SiO_2/Si substrate were obtained for subsequent FET device fabrication. The schematic diagram is presented in Figure S11a.

FET Device Fabrication and Electrical Property Characterization. A fibrous RP nanoribbon-based FET device was fabricated using p-doped silicon substrates covered with a 300 nm thick thermally oxidized silicon dioxide layer. Then, 50 nm thick Au source and drain electrodes were thermally evaporated (MEB-600, Beijing Chuangshiweina Technology Co., LTD.) by using the 600-mesh Cu grid (Gilder Grids) as the physical mask. This facile fabrication technique can greatly alleviate the performance degradation caused by oxidation. The electrical performances were measured by the Agilent B1500 analyzer at room temperature under N_2 atmospheres. Hole mobility is extracted from the linear region of the transfer curves following the equation

$$u = \frac{dI_{ds}}{dV_{gs}} \times \frac{L}{WC_iV_{ds}}$$

where dI_{ds}/dV_g is the slope of the linear regime of transfer curves, C_i is the dielectric capacitance, L and W are the channel length and width between source/drain electrode, respectively, and V_{ds} is the source-drain bias.

DFT Calculations. First-principles computations based on the DFT were performed in the Vienna Ab initio Simulation Package (VASP) code. Electron−electron interactions were treated within the generalized gradient approximation (GGA) proposed by the Perdew−Burke−Ernzerhof (PBE96) standard functional.[42] Electron−ion interactions were described using the projector augmented wave method at a plane-wave cutoff of 500 eV. Lattice constants and atomic positions of bulk fibrous RP were fully relaxed during geometry optimization. Atoms were also allowed to relax sufficiently in 2D slab models of one to four layers, where the vacuum layer is set to be 15 Å to avoid superfluous interactions between periodic replicas. The total energy and the force acting on each atom were converged to 10^{-5} eV and 0.03 eV Å^{-1}, respectively. The 4 × 4 × 7 gamma-centered k-point grids were used for the calculations of the bulk fibrous RP. The 2 × 2 × 1 and 1 × 2 × 4 gamma-centered k-point grids were used for the calculations of the few-layered fibrous RP stacked along the a- and c-axis, respectively. The DFT-D3 method is used to describe the van der Waals interaction throughout the calculations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.chemmater.1c02136.

XRD and Raman spectra of amorphous RP, SEM images, energy-dispersive X-ray and XPS spectra of bulk fibrous RP, calculated electronic structures, UV−vis−NIR diffuse reflectance spectrum, absorption spectra, mechanical exfoliation of fibrous RP, transfer curves of FET, and comparison of FET device performance (PDF)

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Notes

The authors declare no competing financial interest.

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